08-03 第2篇

阅读理解（说明文） How Does a Seed Become a Tree?

① In a small kitchen, a child once dropped an apple seed on the windowsill, put a damp cotton ball on top of it, and waited. Six days later, a tiny white root slipped out of the seed; two weeks later, a green arrow pushed up towards the light. That small quiet scene repeats everywhere on Earth, from the windowsill to the wild forest. A seed — smaller than a coin — holds inside it the complete plan of a tree. But how does a hard, silent seed wake up, break the ground, and become a green living thing? The answer is a short, patient story involving **water, warmth, oxygen, and light**.

② A seed is made of three parts: the **outer coat**, the **embryo** (the baby plant), and the **food store** (often called the "cotyledon"). The outer coat protects the embryo from cold, dryness, and insects — sometimes for years. The embryo is the young plant itself: a tiny stem, a tiny root, and the very first leaves, all folded in like a baby in a blanket. The food store is the lunch-box the mother plant has packed for its child — enough to support the first days of life, before the seedling can feed itself.

③ To wake up, a seed usually needs three things: **water, warmth, and oxygen**. First, water passes through the outer coat and enters the embryo. The hard, dry coat softens, and the embryo swells. Enzymes inside the seed begin to break the food store into smaller, usable pieces — much as saliva in our mouths begins to break bread. Warmth speeds up these chemical steps. Most seeds "wake" best between 15°C and 25°C, which is why gardeners plant in spring rather than winter. Oxygen from the air passes through the outer coat and gives the seed the energy it needs to move its parts. This careful awakening is called **germination**.

④ Once germination begins, the seed does something neat: the **root comes first**. Even if the seed is upside down, the tiny root always grows downwards. Scientists call this response **gravitropism** — the root simply follows the pull of gravity. Meanwhile, the **shoot** (the young stem and first leaves) grows upwards, away from gravity, towards the open air.

⑤ Why does the shoot turn green? Inside its cells are small green machines called **chloroplasts**. Once they reach the light, they begin a process we have all heard of: **photosynthesis**. Using sunlight, water, and the carbon dioxide from the air, chloroplasts build **sugar** — the food the plant now makes for itself. The baby plant has stopped relying on its food store and become, for the rest of its life, a little factory powered by the sun.

⑥ The first true leaves spread out like small solar panels. Over weeks, the root grows longer, pushing deep into the soil, anchoring the tree and drinking water and minerals. The stem thickens and grows up; branches appear. Each year, the tree adds a new layer of wood just beneath the bark, creating the **annual rings** we see on a cut log. A thousand seasons of rain and sun leave a thousand rings of careful growth.

⑦ Not every seed makes it. Many are eaten by birds or squirrels. Others fall in wrong places: on concrete, in deep shade, in water too salty. A wild oak tree may produce ten thousand acorns before one becomes a mature tree. This sounds wasteful, but nature has learned to play a long, patient game. Seeds are cheap to make; trees take years. Producing many seeds keeps the species safe when most of them fail.

⑧ Some seeds have surprising tricks for moving to good places. The seeds of dandelions have tiny parachutes that ride the wind for miles. Maple seeds spin like helicopter blades. Burdock seeds hook onto the fur of animals (an idea that helped a Swiss engineer invent the hook-and-loop fabric we call Velcro). Coconut seeds float across whole oceans before landing on a new beach. Without moving a step, a tree can spread its children to places it will never see.

⑨ So the next time you pass a thick, tall tree, remember: once, it was only a seed smaller than your thumb. It waited, drank water, felt warmth, broke open, stretched towards light — and slowly, over many years, became the quiet green creature that now shades the whole street. Every tall tree is a small seed that chose to keep going. There is something, perhaps, for all of us to learn from that.

根据文章内容，选择最佳答案。

1. What are the three main parts of a seed?

A. The root, the stem, and the flower.
B. The outer coat, the embryo, and the food store (cotyledon).
C. The bark, the trunk, and the branches.
D. The sun, the water, and the soil.

2. According to the passage, why do gardeners plant most seeds in spring?

A. Because spring has the longest nights.
B. Because most seeds "wake" best between 15°C and 25°C, which matches spring warmth.
C. Because seeds hate cool temperatures completely.
D. Because there is no rain in spring.

3. Why does the root grow downwards even when the seed is upside down?

A. Because the soil calls the root.
B. Because of gravitropism — the root follows the pull of gravity.

C. Because of strong wind from above.
D. Because the shoot pushes it downwards.

4. Why does the shoot turn green and make its own food?

A. Because of small green machines (chloroplasts) in its cells, which do photosynthesis using sunlight, water and carbon dioxide.
B. Because the seed is painted green inside.
C. Because the shoot eats small insects near the soil.
D. Because green colour is the only colour plants can see.

5. What does the writer mainly want to say in the last paragraph?

A. Trees are not really important for our lives.
B. Every tall tree is a small seed that chose to keep going, and there is something for us to learn from that quiet patience.
C. Seeds should stay inside the house forever.
D. Only scientists should care about seeds.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段
【解析】外皮、胚、胚乳（子叶）三部分。A/C/D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "best between 15°C and 25°C..."
【解析】春季温度最适宜萌发。A/C/D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "gravitropism — the root simply follows the pull of gravity."
【解析】向地性反应。A/C/D 均错。故选 B。

4.
【答案】A
【题型】细节理解题
【定位】第⑤段
【解析】叶绿体+光合作用。B/C/D 均错。故选 A。

5.
【答案】B
【题型】主旨大意题
【定位】第⑨段末句
【解析】作者借种子谈坚持成长。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **embryo** [n.] 胚
- **cotyledon** [n.] 子叶
- **germination** [n.] 萌发
- **gravitropism** [n.] 向地性
- **chloroplast** [n.] 叶绿体
- **acorn** [n.] 橡子